Leon Sandler
Independent Researcher
Northbrook, Illinois, USA
leon.sandler@alumni.swinburne.edu
ORCID 0009-0007-4584-808X

3 November 2026

The Editor-in-Chief
Waste Management
Elsevier

Dear Editor,

I submit the manuscript “Landfill methane recovery and use under deep uncertainty: a satellite-constrained Monte Carlo decision model for the Norte III landfill, Buenos Aires” as a Research Paper for consideration in Waste Management.

Very large landfills are among the largest anthropogenic methane sources identified from space, yet site decisions are usually made without joining the operator's reported gas capture, independent emission estimates and the economics of the competing uses of the recovered gas. The manuscript does this for the Norte III landfill in Buenos Aires with a Monte Carlo decision model of 300,000 draws. A wide bottom-up generation prior is constrained by the satellite emission estimates at their own observation date, with the capture of that period, and carried forward to the 2025–26 decision period; three observation-error models are compared. The inferred collection efficiency, about 0.13 in 2022 and 0.14 in 2025–26, is reported as a model output conditional on the priors, not as a measurement.

The main results are decision-relevant and robust. None of the eight integrated intervention scenarios has a positive median incremental net present value under the assumed no-credit conditions, although the capture measures abate methane at a median of a few US dollars per tonne of CO_2e, so credit eligibility rather than engineering decides their bankability. Additional engines on genuinely uncommitted surplus gas, reported separately as the competing use, are profitable in 99% of draws at merchant tariffs, whereas biomethane would need a bio-GNC plant-gate price of about US$0.46 per Nm^3 (US$14 per MMBtu). A selective cover reduces net emissions only when the captured share of the gas it blocks exceeds a closed-form break-even (median 0.14), which the simulation reproduces.

The paper states what its central inference does and does not identify. The collection efficiency follows from the satellite emission, the oxidation fraction and the reported capture alone, and does not depend on the generation prior; it is tested against five alternative generation histories, scaling and widening of the satellite estimate, and an independent 2025 observation (Carbon Mapper, 7.6 t CH_4 per hour). That observation is about 29% of what the 2019–22-calibrated model predicts; I report the discrepancy, state that the model is externally unvalidated at facility scale, and show that the decision conclusions do not change across the range of efficiencies (0.14 to 0.24) the two data sets support. The result is conditional on the data and is not a statement about the accuracy of the operator's reported capture. The probability that additional engines are profitable (99% given surplus gas) is separated from the

probability across all operating states (69% after the planned engine expansion), and the climate account of each use is repeated under alternative counterfactuals. The photovoltaic-coating analysis, the full parameter table and the balance checks are in the Supplementary Information, and a field-measurement protocol is proposed that would convert the independent check into a validation.

Three features distinguish the work. Every scenario closes explicit methane, carbon and energy balances, and the climate benefit computed from the balances equals the incremental formulas; the economic priors are anchored to dated Argentine benchmarks for gas, electricity, import duties, country risk, tax and carbon credits; and the conclusions are tested against correlated inputs, alternative and widened priors, structural alternatives for the cover, contracted against merchant electricity, after-tax and WACC discounting, expected credit revenue and a time-varying first-order-decay gas supply. 59 automated tests (mass, carbon and energy closure, units, limiting cases and an independent cash-flow benchmark) must pass before results are produced.

The study is a screening analysis built on published operator figures and satellite estimates; it uses no new site measurements and states which measurements would change the decisions: a calibrated site flux survey, hourly gas and engine dispatch records, and a firm offtake or tariff quotation. The model, tests, parameter table and results are openly available at https://github.com/sandlerleon/norte3-landfill-methane-decision and archived at https://doi.org/10.5281/zenodo.23823890 (all versions); the manuscript is archived at https://doi.org/10.5281/zenodo.23823885 (all versions). One command regenerates every number, table and figure.

A generative AI tool (Claude, Anthropic) assisted with source gathering, code development, drafting and editing; I reviewed and checked all of it and take full responsibility for the content, as disclosed in the manuscript. The manuscript is original, has not been published elsewhere and is not under consideration at any other journal. I am the sole author, I have no competing interests, and the work received no external funding. The photovoltaic-coating concept examined in one scenario originates from my own unpublished competition submission, which is disclosed in the manuscript.

Author affiliation and email address. I am an independent researcher with no current institutional affiliation, and I am the sole author. My contact and corresponding-author address is the alumni email address provided by my former university (Swinburne University of Technology, leon.sandler@alumni.swinburne.edu); it identifies me as a graduate and does not indicate a current affiliation or employer. The manuscript follows the journal's layout requirements: 12 pt type, double spacing, 2.54 cm margins, line and page numbers, and no more than eight figures and tables in the main text, the remaining tables and figures being in the Supplementary Information.

Thank you for considering this submission.

Sincerely,

Leon Sandler